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Research Experience

Search for low-MET electroweak supersymmetry

Texas Tech University

2022-ongoing

- Main topic of the thesis: This analysis focusses on all-hadronic signatures of stealth and RPV decays of charginos and neutralino of the supersymmetry
- The analysis will use CMS collision data from Run 2 (year 2016-2018) and Run 3 (year 2022-onwards)
- These extremely hard to probe models gain sensitivity using novel deep learning jet taggers, increased luminosity and better understanding of the detector
- Events with high energy gluon jets and top pairs are the dominant standard model backgrounds. A combination of particlenet bb, W jet taggers is used to improve sensitivity
- This analysis searches for supersymmetry in previously unexplored region using cutting edge AI and machine learning techniques. If we succeed in observing SUSY signatures, it would represent a major breakthrough in particle physics. If we fail to find any signatures, we will be able to place stronger upper limits on its production cross section.

Jnana Prabodhini Foundation

Pune, India & Lubbock, USA

Research Coordinator

2021

- A 3-year long project aimed at estimating COVID-19 mortality and undercount factor for Pune city in India.
- A frugal method of estimating COVID-19 mortality was tested against traditional statistics-based models widely used in epidemiology and media reports
- The data was collected through public surveying in 2 rounds over a period of a year
- The results were published in nature - scientific reports
- Gained experience in leading a team of researchers and working on topic outside the domain expertise.

Simulating Top Hat Analyzer detector (ASPEX, Aditya-L1)

Physical Research Laboratory, Ahmedabad

Internship

2017-2018

- Simulated the Top Hat Analyzer particle detector, part of the ASPEX payload on ISRO's Aditya-L1 solar mission
- Built a 2D simulation of the detector in SIMION and a detailed 3D model in COMSOL to study particle trajectories and detector response
- Analyzed simulation output in MATLAB/Octave to evaluate detector performance across design variations
- Findings contributed to finalizing the instrument's prototype design

Technical/Service Work

Hadronic Calorimeter (HCAL)

CERN, Geneva

2024

- Member of two international groups responsible for monitoring HCAL operations and performance studies
- Identified and mitigated HCAL operational anomalies: Diagnosed high-frequency "lumi spikes" in the forward HCAL and investigated anomalous peaks in total collected charge to ensure data quality for Run 3.
- Managed detector stability via noise monitoring : Developed and maintained the daily pedestal (noise) trend monitoring framework, directly providing the calibration constants required for downstream event reconstruction.

Light Dark Matter Experiment (LDMX)

Texas Tech University & CERN, Geneva

Trigger Scintillator Detector Lead

2020-2022

- Wrote the trigger scintillator simulation framework for LDMX, simulating multi-particle response and detector performance
- Built and tested a detector prototype at Texas Tech University; constructed a cosmic ray test stand and recorded and analyzed data
- Developed remote control and monitoring software for detector parameters, including stepper-motor-based positioning control software
- Participated in test beam at CERN, where the prototype was tested using high energy pion, electron and muon beams; monitored and controlled collision runs in shifts
- Assumed charge of the trigger scintillator subdetector; results presented at APS conferences

Teaching Experience

Error analysis

Texas Tech University

Lecturer

Aug. 2023 - Nov. 2023

- Delivered lectures on error analysis required in conducting physics experiments
- The class composed of physics and maths majors
- Designed curriculum, weekly asignments and exams
- The lectures comprised of topics on statistics relevent to physics such as error propagation, Chi-square fitting, covariance etc.

Student guidance

Texas Tech University

Mentor

2021 - 2023

- Mentored a student at Stanford University, who took over my work on LDMX trigger scintillator slow control software
- The work was included in the student's Masters thesis
- Mentored another student at Texas Tech University, who analyzed the data recorded at LDMX test beam
- Taught them how the data was recorded, program to analyze the data, features seen in the data
- The student presented the work at meeting of Texas-section of American Physical Society in 2023

Publications

Significant contribution to other areas

- "Conventional and frugal methods of estimating COVID-19-related excess deaths and undercount factors," Dedhe, A., Chowkase, A., Gogate, N., *et al.*, Sci Rep **14**, 10378 (2024) doi: 10.1038/s41598-024-57634-6
- "Current Status and Future Prospects for the Light Dark Matter eXperiment" T. Åkesson, N. Blinov, L. Brand-Baugher, C. Bravo, L. K. Bryngemark, P. Butti, C. Doglioni, C. Dukes, V. Dutta and B. Echenard, *et al.* arXiv:2203.08192

Author on 200+ CMS publications (selected list below)

- "Reweighting simulated events using machine-learning techniques in the CMS experiment", Eur. Phys. J. C **85**, 495 (2025), doi: 10.1140/epjc/s10052-025-14097-x
- "Development of systematic uncertainty-aware neural network trainings for binned-likelihood analyses at the LHC," Eur. Phys. J. C **85**, 1360 (2025) doi:10.1140/epjc/s10052-025-14713-w
- "Determination of the spin and parity of all-charm tetraquarks," Nature **648**, 58-63 (2025) doi:10.1038/s41586-025-09711-7
- "Searches for Higgs boson production through decays of heavy resonances," Phys. Rept. **1115**, 368-447 (2025) doi:10.1016/j.physrep.2024.09.004
- "The CMS Statistical Analysis and Combination Tool: Combine," Comput. Softw. Big Sci. **8**, 19 (2024) doi:10.1007/s41781-024-00121-4
- "Search for Soft Unclustered Energy Patterns in Proton-Proton Collisions at 13 TeV," Phys. Rev. Lett. **133**, 191902 (2024) doi:10.1103/PhysRevLett.133.191902

Workshops and conferences

2026	**Speaker,** APS global physics summit	*Denver, USA*
2024	**Participant,** European School of High Energy Physics	*Peebles, UK*
2024	**Speaker,** APS April Meeting	*Sacramento, USA*
2021	**Speaker,** Meeting of the Division of Particles and Fields of the American Physical Society	*Virtual*
2021	**Speaker,** Joint Meeting of the Texas Sections of APS, AAPT and Zone 13 of the SPS	*Virtual*

Education

Texas Tech University

Lubbock, USA

PhD in Physics

Aug. 2020 - Ongoing

- Currently pursuing PhD, with focus on exerimental high energy physics
- Supervisor: Prof. Sung-won Lee

Texas Tech University

Lubbock, USA

MS in Physics

Aug. 2021

- Masters degree offerred for clearing the PhD candidacy test

Indian Institute of Science Education and Research

Pune, India

BS-MS in Physics

Aug. 2015 - May 2020

- Masters thesis topic: To study charged pion performance in prototype HGCAL detector
- Masters thesis advisor: Prof. Seema Sharma